

## Coding & Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 15 July 2026                    |
| <b>Team ID</b>       | xxxxxx                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                    | Details                                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL    | <a href="https://github.com/KammarapalliMeghana/CreditCardApprovalPrediction">https://github.com/KammarapalliMeghana/CreditCardApprovalPrediction</a>                         |
| Programming Language(s)  | Python, HTML, CSS, JavaScript                                                                                                                                                 |
| Framework(s) Used        | Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn                                                                                                                       |
| Key Features Implemented | • Credit card approval prediction using Machine Learning. • Data preprocessing and feature engineering. • Data visualization and exploratory analysis. • Training of Logistic |

| Field                         | Details                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Regression, Decision Tree, Random Forest, and XGBoost models. • Interactive Flask web application for real-time prediction. • Deployment on Render.                                                                                                                                                                                                                                                                   |
| Pending / Incomplete Features | <ul style="list-style-type: none"> <li>• User authentication system.</li> <li>• Database integration for storing prediction history.</li> <li>• Email notification feature.</li> <li>• Advanced dashboard and analytics.</li> </ul>                                                                                                                                                                                   |
| Setup / Run Instructions      | <b>1. Clone Repository:</b> git clone <a href="https://github.com/KammarapalliMeghana/CreditCardApprovalPredicton.git">https://github.com/KammarapalliMeghana/CreditCardApprovalPredicton.git</a><br><b>2. Install dependencies:</b> pip install -r requirements.txt<br><b>3. Train model:</b> python model_training.py<br><b>4. Run application:</b> python app.py<br><b>5. Open browser:</b> http://127.0.0.1:5000/ |



## Code Quality Checklist

| S.No | Criteria                                              | Status (Yes / No) |
|------|-------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions/classes  | Yes               |
| 2    | Meaningful variable and function names are used       | Yes               |
| 3    | Code includes comments/documentation where necessary  | Yes               |
| 4    | Error handling is implemented for critical operations | Yes               |
| 5    | The application runs without critical errors          | Yes               |
| 6    | Code is committed to a version control repository     | Yes               |

## Additional Notes / Comments:

The project follows a modular structure with separate folders for datasets, templates, static files, and model files.

Multiple machine learning algorithms were implemented and compared to select the best-performing model.

The application provides real-time credit card approval prediction through a user-friendly web interface.

The trained model achieved approximately **99–100% accuracy** on the generated dataset.